

Section A

Answer **all** the questions in this section.

Question 1 [11 marks]

Fig. 1.1 shows two neurons in contact with one another.

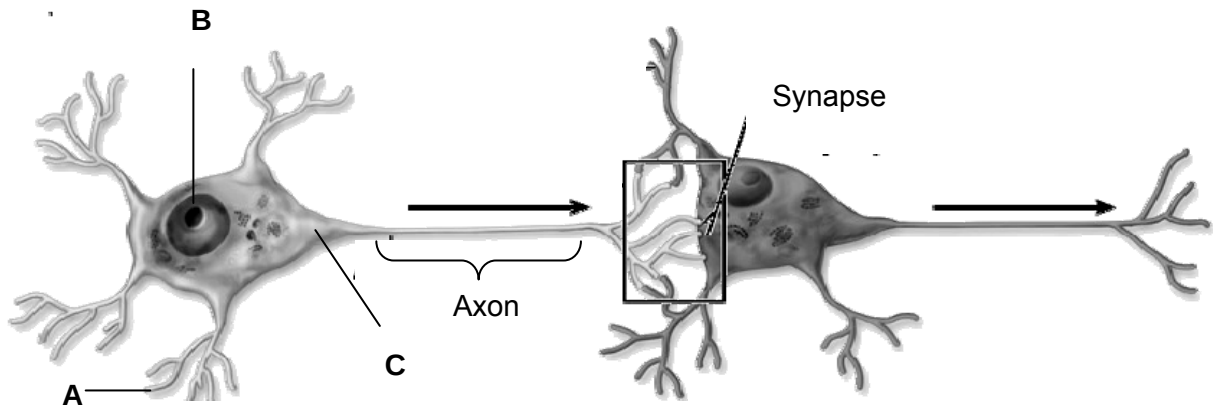


Fig. 1.1

(a) Identify the parts labeled **A**, **B** and **C**.

A: _____

B: _____

C: _____ [3]

(b) Explain the role of membrane proteins in maintaining resting potential of the axon (-70mv).

[4]

(c) Fig. 1.2 below shows a detailed diagram of a cholinergic synapse.

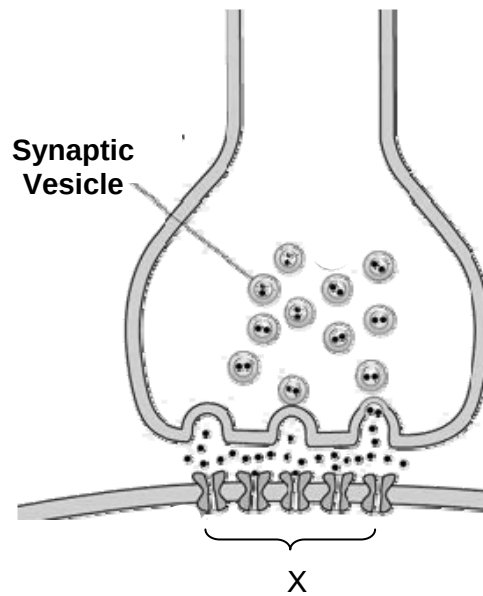


Fig. 1.2

(i) Describe the role of X in a cholinergic synapse.

[2]

The black widow spider venom contains a toxin called latrotoxin that affects the normal functioning of the synapse. The bite of the black widow spider initially produces an increase in neuromuscular activity that leads to painful skeletal muscle spasms and rigidity. This phase of hyperexcitability is rapidly followed by progressive failure of neuromuscular transmission and paralysis. Experiments have found that latrotoxin can form permeable calcium ion pores on the presynaptic membrane.

(d) Suggest what happens at the synapse of a person bitten by a black widow spider.

[2]

Question 2 [10 marks]

ATP synthase, also known as a stalked particle, is an enzyme that synthesizes ATP from ADP and inorganic phosphate (P_i). Fig. 2.1 shows the structure of ATP synthase. It is found in both the chloroplast and the mitochondria.

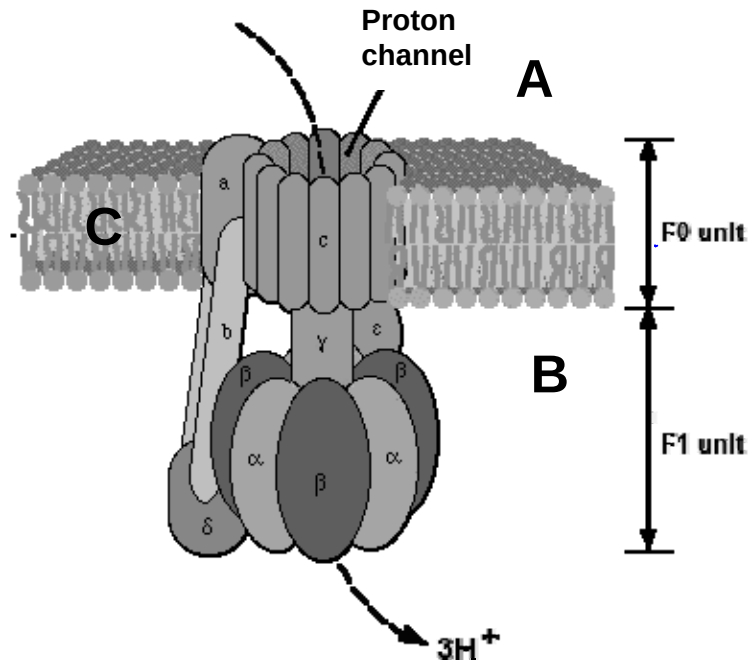


Fig. 2.1

(a) Identify the regions (A, B, C) as found in the mitochondrion and chloroplast respectively.

	A	B	C
Mitochondrion			
Chloroplast			

[3]

In both organelles, a proton gradient exists across the membrane where ATP synthase is located. The synthesis of ATP by ATP synthase is coupled with the diffusion of H^+ across the proton channel down its concentration gradient.

(b) Describe how this proton gradient is generated in

(i) the mitochondria.

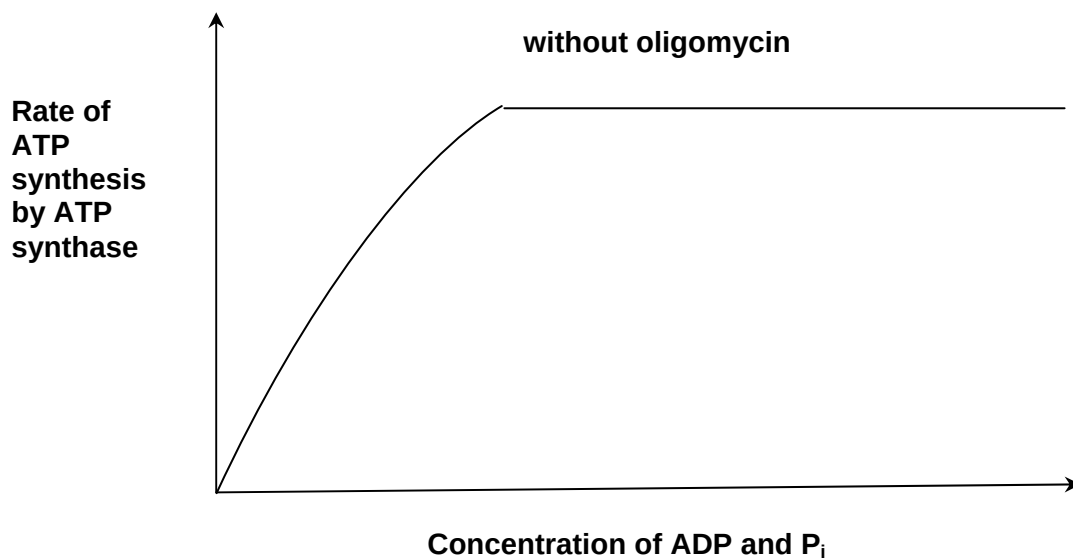
[3]

(ii) the chloroplast.

[3]

Oligomycin is a chemical that inhibits ATP synthase by blocking the proton channel. When the proton channel is blocked, H^+ no longer diffuses across the membrane and thus ATP synthesis stops. Administering oligomycin to an individual will result in high levels of lactate in blood and urine.

(c) Indicate in the graph below, the effect of administering a fixed amount of oligomycin on the rate of ATP synthesis.



[1]

Question 3 [14 marks]

Fig. 3.1 is an electron micrograph of a eukaryotic interphase nucleus.

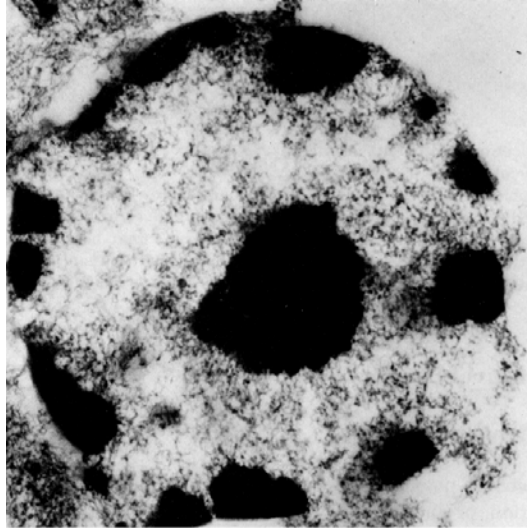


Fig. 3.1

(a) Describe the structural organization of DNA in a eukaryotic interphase nucleus.

[4]

(b) Explain why changes to DNA structure are required before transcription can take place.

[2]

In eukaryotic transcription apart from structural changes, control elements also play a major role in controlling gene expression at the transcriptional level.

- (c) Explain the role of **two** control elements that are involved in influencing eukaryotic transcription.

[4]

Fig. 3.2 shows a protein which plays a central role in eukaryotic transcription. The binding of this protein forms the first step in the formation of the transcription initiation complex.

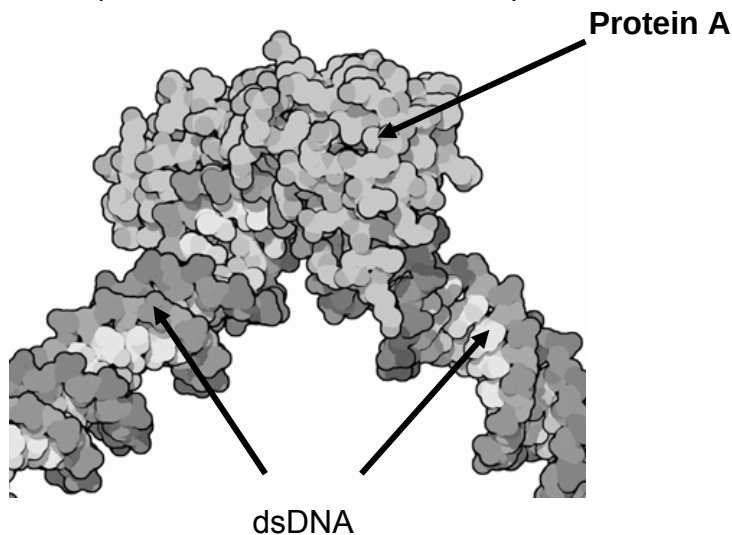


Fig. 3.2

- (d) Based on Fig 3.2, precisely identify **Protein A** and the region on the DNA that **Protein A** binds.

[2]

- (e) Suggest how **Protein A** associates with the DNA.

[2]

Question 4 [14 marks]

One of the key differences in protein synthesis between prokaryotes and eukaryotes is that eukaryotic cells perform various post-transcriptional modifications of their RNA molecules. Capping and polyadenylation are two kinds of post-transcriptional modifications.

(a) Describe briefly what is meant by capping and polyadenylation.

[2]

In an experiment to investigate the function(s) of capping and polyadenylation, scientists introduced firefly luciferase mRNA, with and without a cap, and with and without poly(A), into tobacco cells. The amount of remaining luciferase mRNA and the luciferase activity were measured experimentally. Luciferase is a protein product that is easy to detect because of the light it generates in the presence of appropriate substances (luciferin and ATP). Luciferase activity would therefore be a reflection of the amount of proteins produced (translation efficiency)

Table 4.1 below illustrates the effects of capping and polyadenylation on the lifespan of the luciferase mRNA and its protein products.

Table 4.1

mRNA	Luciferase mRNA Half Life (min)	Luciferase Activity (units)
Uncapped		
Poly(A) tail absent	31	2,941
Poly(A) tail present	44	4,480
Capped		
Poly(A) tail absent	53	62,595
Poly(A) tail present	100	1,331,917

Note : Half life refers to the time taken for the amount of mRNA to be degraded to half the original amount

(b) With reference to the Table 4.1, describe the effect of polyadenylation on mRNA degradation in capped mRNA.

[2]

- (c) Does capping or polyadenylation have a greater impact on translation efficiency?
Explain your answer using data from Table 4.1.

[3]

- (d) Briefly describe another post-transcriptional modification which occurs in eukaryotic cells.

[2]

Prokaryotic mRNA is very short lived. In *E. coli* the average half life of some mRNAs is only about two minutes. This is because prokaryotic mRNAs are not stabilized by post-transcriptional modifications. Therefore there is a very short time interval between transcription and translation as the mRNAs are constantly broken down into their constituent ribonucleotides by ribonucleases.

- (e) Explain why translation of each prokaryotic mRNA will still always take place in spite of the rapid degradation of mRNA.

[2]

- (f) Apart from the differences due to post-transcriptional modifications,

- (i) describe one other structural difference between prokaryotic and eukaryotic mRNA.

[2]

- (ii) explain how the structural difference stated in f(i) arises.

[1]

Question 5 [10 marks]

Fig. 5.1 shows the structure of an influenza virus.

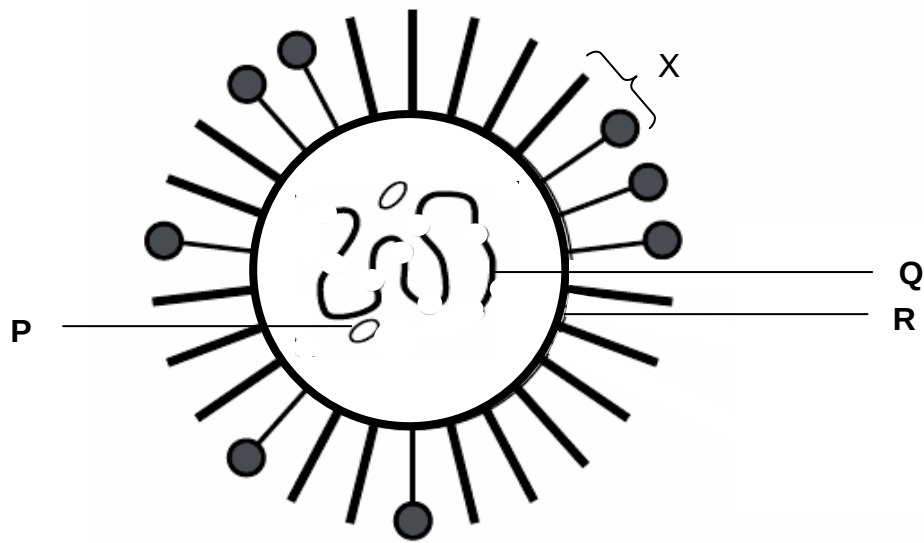


Fig. 5.1

(a) Identify the parts labeled **P**, **Q** and **R**.

P: _____

Q: _____

R: _____ [3]

(b) With reference to Fig. 5.1, briefly explain

(i) the function of **P**.

 _____ [1]

(ii) how **X** is synthesized.

 _____ [3]

(c) Describe how the influenza virus enters the host cell.

[2]

Three distinct types of influenza virus, type **A**, **B**, and **C**, have been identified. The most common human vaccine is the trivalent influenza vaccine that contains purified and inactivated material from three viral strains. A vaccine formulated for one year may be ineffective in the following year.

(d) Suggest why “a vaccine formulated for one year may be ineffective in the **following year**”.

[1]

Question 6 [11 marks]

The common fruit fly, *Drosophila melanogaster*, found in Australia is an economic pest. In a population of flies studied, there are flies with black body and normal wings. There are also flies with white body and vestigial wings.

The allele for black body was assumed to be dominant to the allele for white body (absence of pigment). The allele for normal wings was also assumed to be dominant to the allele for vestigial wings.

- (a) Assuming the alleles of the two genes segregate independently, state the offspring phenotypes and the respective phenotypic ratio of a cross between two heterozygous flies.

[2]

When an actual cross between a homozygous black body fly with normal wings and a homozygous white body fly with vestigial wings was carried out, all offspring have grey body and normal wings.

- (b) It was stated that the allele for black body is dominant over the allele for white body colour. State if this assumption is valid. Explain your answer.

[2]

In addition, when the grey body normal wing offspring is mated with an individual with white body and vestigial wings, the following results are obtained.

grey body and normal wings	545
grey body and vestigial wings	109
white body and normal wings	97
white body and vestigial wings	532

Use the following symbols:

B represents the allele for black body

b represents the allele for white body

N represents the allele for normal wings

n represents the allele for vestigial wings

(c) Use a genetic diagram to explain the cross above.

- (d) A chi-square test was performed on these data giving a calculated χ^2 of 15.37. Use the calculated value of χ^2 and the χ^2 table provided in Table 6.1 to draw conclusion on the validity of the hypothesis that the two genes segregate independently.

Table 6.1

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Conclusion:

[2]

Question 7 [10 marks]

Taxonomists organize species into hierarchical systems and thereby attempt to show evolutionary relationships. The algae have proved difficult to classify with other organisms, and they are now placed with the protozoans (kingdom Protoctista), in a separate kingdom from land plants. All the organisms in this kingdom are eukaryotes, but apart from this they share very few common features.

5

Among the algae there is a wide variation in vegetative morphology. Many of the free-floating microscopic species that form the phytoplankton in freshwater and in the sea are unicellular. Some species are loose associations or colonies of similar cells. The larger algae are multicellular and are either filamentous or thalloid. In the more complex forms such as the large kelps, there is some differentiation into tissues, but there are no organs. This range of algal structure is useful in designing keys for identification, but is of little value in devising natural classification systems.

10

The photosynthetic pigments found in algae are used by taxonomists to distinguish between the major groups or phyla. These phyla are the green algae (Chlorophyta), the brown algae (Phaeophyta) and the red algae (Rhodophyta). Although they all show similar variation in vegetative morphology, these phyla are not thought to be closely related. The major features of these phyla are shown in Table 7.1.

15

Table 7.1

Features	Chlorophyta	Phaeophyta	Rhodophyta
Vegetative morphology	Unicellular, colonial, thalloid	Simple or branched, filamentous, thalloid	Unicellular filamentous, thalloid
Flagellate stages in the life cycle	✓	✓	X
Carbohydrate storage product	Starch	Laminarin	Floridean starch
Cell wall polysaccharides	Cellulose	Cellulose, alginic acid, fucinic acid	Cellulose, polysulphate, esters
Photosynthetic pigments:			
Chlorophyll a	✓	✓	✓
Chlorophyll b	✓	X	X
B carotene	✓	✓	✓
Lutein	✓	X	✓
Fucoxanthin	X	✓	X
R-phycoerythrin	X	X	✓
R-phyococyanin	X	X	✓

(a) With reference to Table 7.1, explain what is meant by wide variation in vegetative morphology (line 6).

[1]

Fig. 7.1 is a cladogram that was constructed to represent the relationships between the three phyla of algae based on three features obtained from Table 7.1.

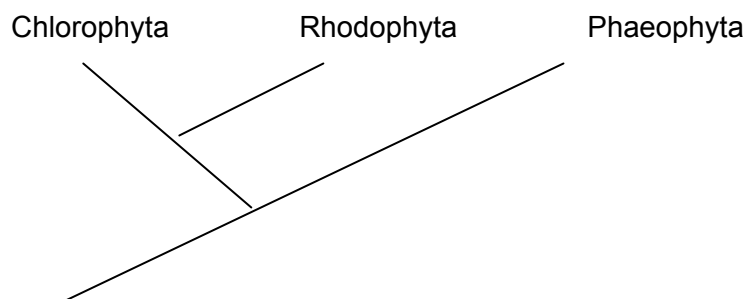


Fig. 7.1

(b) Identify **two** of the three features used in constructing the cladogram in Fig. 7.1.

[2]

(c) Suggest **one** problem, of a practical nature, that may be encountered by a taxonomist who is studying the life history of an alga.

[1]

The mitochondrial DNA (mtDNA) of a red alga, *Porphyra purpurea*, a circular genome of 36.753 kbp, has been completely sequenced and a total of 57 densely packed genes have been identified. Comparisons were made with mtDNAs from animals, fungi, protists and plants, and there is strong evidence that red algae share a common ancestry with green algae and plants. Fig. 7.2 shows a phylogenetic tree constructed from these studies.

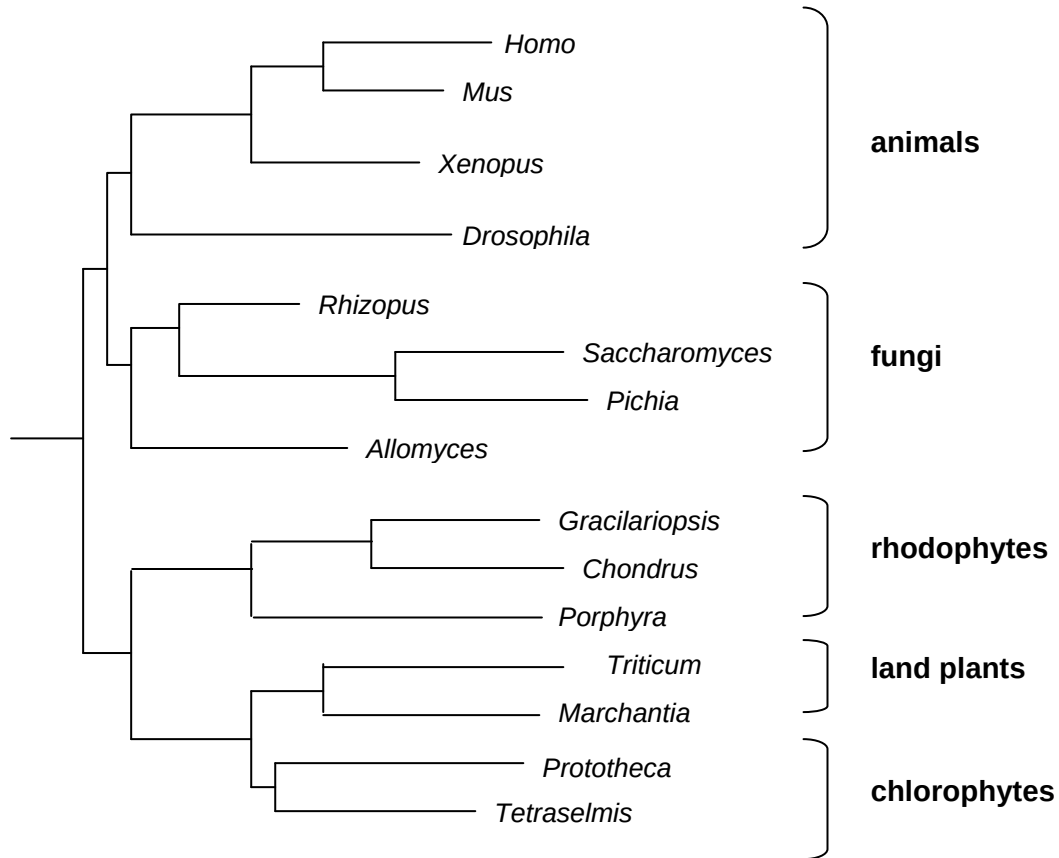


Fig. 7.2

(d) With reference to Fig. 7.2, comment on the relationship between green and red alga as compared to land plants.

[1]

(e) With reference to the relationship between rhodophytes and chlorophytes, explain why different results were obtained from the comparison of major features and from molecular studies.

[3]

- (f) State the difference between the molecular method used in the study of red alga and the DNA hybridization method.

[2]

End of Section A

Section B

Answer **one** question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a) Outline the main differences between a fibrous protein and DNA. [6]
- (b) Describe three different species concepts and their limitations. [9]
- (c) With reference to the Islets of Langerhans, describe what is meant by an endocrine gland. [5]
- 9 (a) Explain how changes in the structure of chromosomes lead to the development of cancer. [6]
- (b) Explain how meiosis results in genetic variation in gametes. [6]
- (c) Explain the role of F plasmids in conjugation. [8]

End of Paper 2